CHAPTER C10
ICE LOADS—ATMOSPHERIC ICING

C10.1 GENERAL

In most of the contiguous United States, freezing rain is considered the cause of the most severe ice loads. Values for ice thicknesses caused by in-cloud icing and snow suitable for inclusion in this standard are not currently available.

Very few sources of direct information or observations of naturally occurring ice accretions (of any type) are available. Bennett (1959) presents the geographical distribution of the occurrence of ice on utility wires from data compiled by various railroad, electric power, and telephone associations in the nine-year period from the winter of 1928–1929 to the winter of 1936–1937. The data include measurements of all forms of ice accretion on wires, including glaze ice, rime ice, and accreted snow, but do not differentiate among them. Ice thicknesses were measured on wires of various diameters, heights above ground, and exposures. No standardized technique was used in measuring the thickness. The maximum ice thickness observed during the nine-year period in each of 975 squares, 60 mi (97 km) on a side, in a grid covering the contiguous United States is reported. In every state except Florida, thickness measurements of accretions with unknown densities of approximately one radial inch were reported. Information on the geographical distribution of the number of storms in this nine-year period with ice accretions greater than specified thicknesses is also included.

Tattelman and Gringorten (1973) reviewed ice load data, storm descriptions, and damage estimates in several meteorological publications to estimate maximum ice thicknesses with a 50-year mean recurrence interval in each of seven regions in the United States. *Storm Data* (NOAA 1959–Present) is a monthly publication that describes damage from storms of all sorts throughout the United States. The compilation of this qualitative information on storms causing damaging ice accretions in a particular region can be used to estimate the severity of ice and wind-on-ice loads. The Electric Power Research Institute has compiled a database of icing events from the reports in *Storm Data* (Shan and Marr 1996). Damage severity maps were also prepared.

Bernstein and Brown (1997) and Robbins and Cortinas (1996) provide information on freezing rain climatology for the 48 contiguous states based on recent meteorological data.

C10.1.1 Site-Specific Studies. In-cloud icing may cause significant loadings on ice-sensitive structures in mountainous regions and for very tall structures in other areas. Mulherin (1996) reports that of 120 communications tower failures in the United States caused by atmospheric icing, 38 were caused by in-cloud icing, and in-cloud icing combined with freezing rain caused an additional 26 failures. In-cloud ice accretion is very sensitive to the degree of exposure to moisture-laden clouds, which is related to terrain, elevation, and wind direction and velocity. Large differences in accretion size can occur over a few hundred feet and can cause severe load unbalances in overhead wire systems. Advice from a meteorologist familiar with the area is particularly valuable in these circumstances. In Arizona, New Mexico, and the panhandles of Texas and Oklahoma, the United States Forest Service specifies ice loads caused by in-cloud icing for towers constructed at specific mountaintop sites (U.S. Forest Service 1994). Severe in-cloud icing has been observed in southern California (Mallory and Leavengood 1983a, 1983b), eastern Colorado (NOAA Feb. 1978), the Pacific Northwest (Winkleman 1974; Richmond et al. 1977; Sinclair and Thorkildson 1980), Alaska (Ryerson and Claffey 1991), and the Appalachians (Ryerson 1987, 1988a, 1988b, 1990; Govoni 1990).

Snow accretions also can result in severe structural loads and may occur anywhere snow falls, even in localities that may experience only one or two snow events per year. Some examples of locations where snow accretion events resulted in significant damage to structures are Nebraska (NPPD 1976), Maryland (Mozer and West 1983), Pennsylvania (Goodwin et al. 1983), Georgia and North Carolina (Lott 1993), Colorado (McCormick and Pohlman 1993), Alaska (Peabody and Wyman 2005), and the Pacific Northwest (Hall 1977; Richmond et al. 1977).

For Alaska, available information indicates that moderate to severe snow and in-cloud icing can be expected. The measurements made by Golden Valley Electric Association (Jones et al. 2002) are consistent in magnitude with visual observations across a broad area of central Alaska (Peabody 1993). Several meteorological studies using an ice accretion model to estimate ice loads have been performed for high-voltage transmission lines in Alaska (Gouze and Richmond 1982a, 1982b; Richmond 1985, 1991, 1992; Peterka et al. 1996). Estimated 50-year mean recurrence interval accretion thicknesses from snow range from 1.0 to 5.5 in. (25 to 140 mm), and in-cloud ice accretions range from 0.5 to 6.0 in. (12 to 150 mm). The assumed accretion densities for snow and in-cloud ice accretions, respectively, were 5 to 31 lb/ft^3 (80 to 500 kg/m^3) and 25 lb/ft^3 (400 kg/m^3). These loads are valid only for the particular regions studied and are highly dependent on the elevation and local terrain features.

In Hawaii, for areas where freezing rain (Wylie 1958), snow, and in-cloud icing are known to occur at higher elevations, site-specific meteorological investigations are needed.

Local records and experience should be considered when establishing the design ice thickness, concurrent wind speed, and concurrent temperature. In determining equivalent radial ice thicknesses from historical weather data, the quality, completeness, and accuracy of the data should be considered along with the robustness of the ice accretion algorithm. Meteorological stations may be closed by ice storms because of power outages, anemometers may be iced over, and hourly precipitation data recorded only after the storm when the ice in the rain gauge melts. These problems are likely to be more severe at automatic weather stations where observers are not available to estimate the weather parameters or correct erroneous readings. Note also that (1) air temperatures are recorded only to the nearest 1°F, at best, and may vary significantly from the recorded value in the region around the weather station; (2) the wind speed during freezing rain has a significant effect on the accreted ice load on objects oriented perpendicular to the wind direction; (3) wind speed and direction vary with terrain and exposure; (4) enhanced precipitation may occur on the windward side of mountainous terrain; and (5) ice may remain on the structure for days or weeks after freezing rain ends, subjecting the iced structure to wind speeds that may be significantly higher than those that accompanied the freezing rain. These factors should be considered both in estimating the accreted ice thickness at a weather station in past storms and in extrapolating those thicknesses to a specific site.

In using local data, it must also be emphasized that sampling errors can lead to large uncertainties in the specification of the 500-year ice thickness. Sampling errors are the errors associated with the limited size of the climatological data samples (years of record). When local records of limited extent are used to determine extreme ice thicknesses, care should be exercised in their use.

A robust ice accretion algorithm is not sensitive to small changes in input variables. For example, because temperatures are normally recorded in whole degrees, the calculated amount of ice accreted should not be sensitive to temperature changes of fractions of a degree.

C10.1.2 Dynamic Loads. While design for dynamic loads is not specifically addressed in this edition of the standard, the effects of dynamic loads are an important consideration for some ice-sensitive structures and should be considered in the design when they are anticipated to be significant. For example, large-amplitude galloping (Rawlins 1979; Section 6.2 of Simiu and Scanlan 1996) of guys and overhead cable systems occurs in many areas. The motion of the cables can cause damage because of direct impact of the cables on other cables or structures and can also cause damage because of wear and fatigue of the cables and other components of the structure (White 1999). Ice shedding from the guys on guyed masts can cause substantial dynamic loads in the mast.

C10.1.3 Exclusions. Additional guidance is available in Committee on Electrical Transmission Structures (1982) and CSA (1987, 1994).

C10.2 DEFINITIONS

FREEZING RAIN: Freezing rain occurs when warm, moist air is forced over a layer of subfreezing air at the Earth's surface. The precipitation usually begins as snow that melts as it falls through the layer of warm air aloft. The drops then cool as they fall through the cold surface air layer and freeze on contact with structures or the ground. Upper air data indicate that the cold surface air layer is typically between 1,000 and 3,900 ft (300 and 1,200 m) thick (Young 1978), averaging 1,600 ft (500 m) (Bocchieri 1980). The warm air layer aloft averages 5,000 ft (1,500 m) thick in freezing rain, but in freezing drizzle the entire temperature profile may be below 32°F (0°C) (Bocchieri 1980).

Precipitation rates and wind speeds are typically low to moderate in freezing rainstorms. In freezing rain, the water impingement rate is often greater than the freezing rate. The excess water drips off and may freeze as icicles, resulting in a variety of accretion shapes that range from a smooth cylindrical

FIGURE C10.2-1 Glaze Ice Accretion Caused by Freezing Rain

sheath, through a crescent on the windward side with icicles hanging on the bottom, to large irregular protuberances, see Fig. C10.2-1. The shape of an accretion depends on a combination of varying meteorological factors and the cross-sectional shape of the structural member, its spatial orientation, and flexibility.

Note that the theoretical maximum density of ice (917 kg/m^3 or 57 lb/ft^3) is never reached in naturally formed accretions because of the presence of air bubbles.

HOARFROST: Hoarfrost, which is often confused with rime, forms by a completely different process. Hoarfrost is an accumulation of ice crystals formed by direct deposition of water vapor from the air on an exposed object. Because it forms on objects with surface temperatures that have fallen below the frost point (a dew point temperature below freezing) of the surrounding air because of strong radiational cooling, hoarfrost is often found early in the morning after a clear, cold night. It is feathery in appearance and typically accretes up to about 1 in. (25 mm) in thickness with very little weight. Hoarfrost does not constitute a significant loading problem; however, it is a very good collector of supercooled fog droplets. In light winds, a hoarfrost-coated wire may accrete rime faster than a bare wire (Power 1983).

ICE-SENSITIVE STRUCTURES: Ice-sensitive structures are structures for which the load effects from atmospheric icing control the design of part or all of the structural system. Many open structures are efficient ice collectors, so ice accretions can have a significant load effect. The sensitivity of an open structure to ice loads depends on the size and number of structural members, components, and appurtenances and also on the other loads for which the structure is designed. For example, the additional weight of ice that may accrete on a heavy wide-flange member is smaller in proportion to the dead load than the same ice thickness on a light angle member. Also, the percentage increase in projected area for wind loads is smaller for the wide-flange member than for the angle member. For some open structures, other design loads, for example, snow loads and live loads on a catwalk floor, may be larger than the design ice load.

IN-CLOUD ICING: This icing condition occurs when a cloud or fog (consisting of supercooled water droplets 100 microns (100 mm) or less in diameter) encounters a surface that is at or below freezing temperature. It occurs in mountainous areas where adiabatic cooling causes saturation of the atmosphere to occur at temperatures below freezing, in free air in supercooled clouds, and in supercooled fogs produced by a stable air mass with a strong temperature inversion. In-cloud ice accretions can reach thicknesses of 1 ft (0.30 m) or more because the icing conditions can include high winds and typically persist or recur episodically during long periods of subfreezing temperatures. Large concentrations of supercooled droplets are not common at air temperatures below about 0°F (−18°C).

In-cloud ice accretions have densities ranging from that of low-density rime to glaze. When convective and evaporative cooling removes the heat of fusion as fast as it is released by the freezing droplets, the drops freeze on impact. When the cooling rate is lower, the droplets do not completely freeze on impact. The unfrozen water then spreads out on the object and may flow completely around it and even drip off to form icicles. The degree to which the droplets spread as they collide with the structure and freeze governs how much air is incorporated in the accretion, and thus its density. The density of ice accretions caused by in-cloud icing varies over a wide range from 5 to 56 pcf (80 to 900 $\mathrm{kg/m^3}$) (Macklin 1962; Jones 1990). The resulting accretion can be either white or clear, possibly with attached icicles; see Fig. C10.2-2.

The amount of ice accreted during in-cloud icing depends on the size of the accreting object, the duration of the icing condition, and the wind speed. If, as often occurs, wind speed increases and air temperature decreases with height above ground, larger amounts of ice accrete on taller structures. The accretion shape depends on the flexibility of the structural member, component, or appurtenance. If it is free to rotate, such as a long guy or a long span of a single conductor or wire, the ice accretes with a roughly circular cross section. On more rigid structural members, components, and appurtenances, the ice forms in irregular pennant shapes extending into the wind.

SNOW: Under certain conditions, snow falling on objects may adhere because of capillary forces, interparticle freezing (Colbeck and Ackley 1982), and/or sintering (Kuroiwa 1962). On objects with circular cross section, such as a wire, cable, conductor, or guy, sliding, deformation, and/or torsional rotation of the underlying cable may occur, resulting in the formation of a cylindrical sleeve, even around bundled conductors and wires; see Fig. C10.2-3. Because accreting snow is often accompanied by high winds, the density of accretions may be much higher than the density of the same snowfall on the ground.

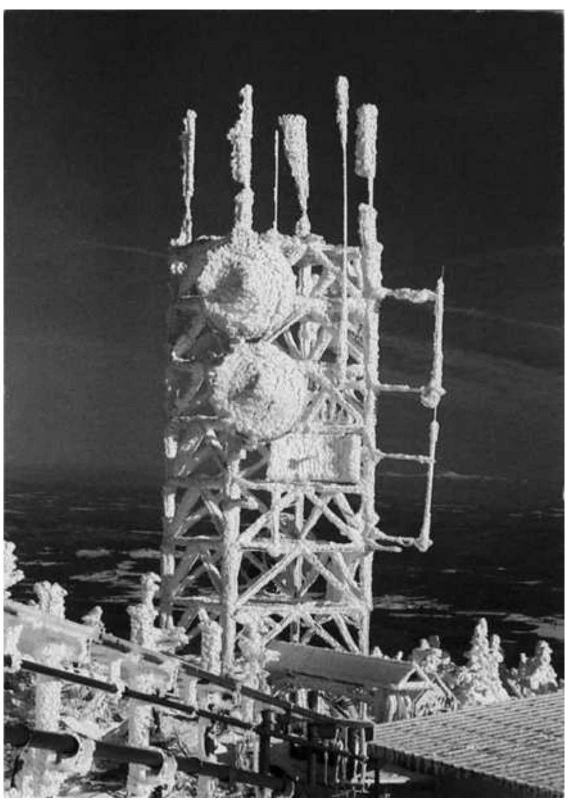

FIGURE C10.2-2 Rime Ice Accretion Caused by In-Cloud Icing

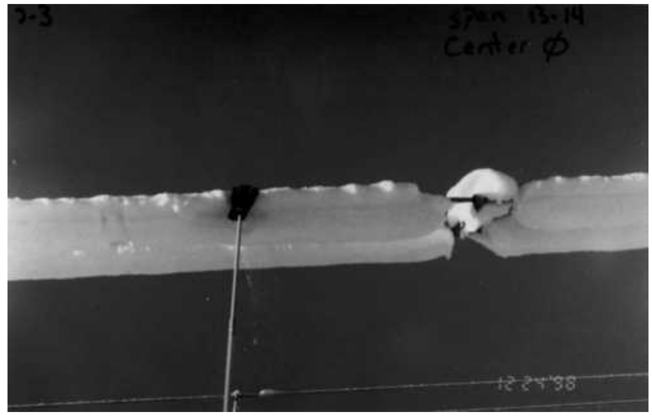

FIGURE C10.2-3 Snow Accretion on Wires

Damaging snow accretions have been observed at surface air temperatures ranging from about 23 to 36°F (−5 to 2°C). Snow with a high moisture content appears to stick more readily than drier snow. Snow falling at a surface air temperature above freezing may accrete even at wind speeds above 25 mi/h (10 m/s), producing dense 37 to 50 pcf (600 to 800 $\mathrm{kg/m^3}$) accretions. Snow with a lower moisture content is not as sticky, blowing off the structure in high winds. These accreted snow densities are typically between 2.5 and 16 pcf (40 and 250 $\mathrm{kg/m^3}$) (Kuroiwa 1965).

Even apparently dry snow can accrete on structures (Gland and Admirat 1986). The cohesive strength of the dry snow is initially supplied by the interlocking of the flakes and ultimately by sintering, as molecular diffusion increases the bond area between adjacent snowflakes. These dry snow accretions appear to form only in very low winds and have densities estimated at between 5 and 10 pcf (80 and 150 $\mathrm{kg/m^3}$) (Sakamoto et al. 1990; Peabody 1993).

C10.4 ICE LOADS CAUSED BY FREEZING RAIN

C10.4.1 Ice Weight. The ice thicknesses shown in Figs. 10.4-2 through 10.4-6 were determined for a horizontal cylinder oriented perpendicular to the wind. These ice thicknesses cannot be applied directly to cross sections that are not round, such as channels and angles. However, the ice area from Eq. (10.4-1) is the same for all shapes for which the circumscribed circles have equal diameters (Peabody and Jones 2002; Jones and Peabody 2006). It is assumed that the maximum dimension of the cross section is perpendicular to the trajectory of the raindrops. Similarly, the ice volume in Eq. (10.4-2) is for a flat plate perpendicular to the trajectory of the raindrops. The constant π in Eq. (10.4-2) corrects the thickness from that on a cylinder to the thickness on a flat plate. For vertical cylinders and horizontal cylinders parallel to the wind direction, the ice area given by Eq. (10.4-1) is conservative.

C10.4.2 Nominal Ice Thickness. The 500-year mean recurrence interval ice thicknesses shown in Figs. 10.4-2 to 10.4-6 are based on studies using an ice accretion model and local data.

FIGURE C10.4-1 Locations of Weather Stations Used in Preparation of Figures 10.4-2 through 10.4-6

Historical weather data from 540 National Weather Service (NWS), military, Federal Aviation Administration (FAA), and Environment Canada weather stations were used with the U.S. Army's Cold Regions Research and Engineering Laboratory (CRREL) and simple ice accretion models (Jones 1996, 1998) to estimate uniform radial glaze ice thicknesses in past freezing rainstorms. The models and algorithms have been applied to additional stations in Canada along the border of the lower 48 states. The station locations are shown in Fig. C10.4-1 for the 48 contiguous states and in Fig. 10.4-6 for Alaska. The period of record of the meteorological data at any station is typically 20 to 50 years. The ice accretion models use weather and precipitation data to simulate the accretion of ice on cylinders 33 ft (10 m) above the ground, oriented perpendicular to the wind direction in freezing rainstorms. Accreted ice is assumed to remain on the cylinder until after freezing rain ceases and the air temperature increases to at least 33°F (0.6°C). At each station, the maximum ice thickness and the maximum wind-on-ice load were determined for each storm. Severe storms, those with significant ice or wind-on-ice loads at one or more weather stations, were researched in *Storm Data* (NOAA 1959–Present), newspapers, and utility reports to obtain corroborating qualitative information on the extent of and damage from the storm. Yet very little corroborating information was obtained about damaging freezing rainstorms in Alaska, perhaps because of the low population density and relatively sparse newspaper coverage in the state.

Extreme ice thicknesses were determined from an extreme value analysis using the peaks-over-threshold method and the generalized Pareto distribution (Hoskings and Wallis 1987; Wang 1991; Abild et al. 1992). To reduce sampling error, weather stations were grouped into superstations (Peterka 1992) based on the incidence of severe storms, the frequency of freezing rainstorms, latitude, proximity to large bodies of water, elevation, and terrain. Concurrent wind-on-ice speeds were back-calculated from the extreme wind-on-ice load and the extreme ice thickness. The analysis of the weather data and the calculation of extreme ice thicknesses are described in more detail in Jones et al. (2002).

The maps in Figs. 10.4-2 to 10.4-6 represents the most consistent and best available nationwide map for nominal design ice thicknesses and wind-on-ice speeds. The icing model used to produce the map has not, however, been verified with a large set of collocated measurements of meteorological data and uniform radial ice thicknesses. Furthermore, the weather stations used to develop this map are almost all at airports. Structures in more exposed locations at higher elevations or in valleys or gorges, for example, Signal and Lookout Mountains in Tennessee, the Pontotoc Ridge and the edge of the Yazoo Basin in Mississippi, the Shenandoah Valley and Poor Mountain in Virginia, Mount Washington in New Hampshire, and Buffalo Ridge in Minnesota and South Dakota, may be subject to larger ice thicknesses and higher concurrent wind speeds. However, structures in more sheltered locations, for example, along the north shore of Lake Superior within 300 vertical ft (90 m) of the lake, may be subject to smaller ice thicknesses and lower concurrent wind speeds. Loads from snow or in-cloud icing may be more severe than those from freezing rain (see Section C10.1.1).

Special Icing Regions. Special icing regions are identified on the map. As described previously, freezing rain occurs only under special conditions when a cold, relatively shallow layer of air at the surface is overrun by warm, moist air aloft. For this reason, severe freezing rainstorms at high elevations in mountainous terrain typically do not occur in the same weather systems that cause severe freezing rainstorms at the nearest airport with a weather station. Furthermore, in these regions, ice thicknesses and wind-on-ice loads may vary significantly over short distances because of local variations in elevation, topography, and exposure. In these mountainous regions, the values given in

Fig. 10.4-1 should be adjusted, based on local historical records and experience, to account for possibly higher ice loads from both freezing rain and in-cloud icing (see Section C10.1.1).

C10.4.4 Importance Factors. The importance factors for ice and concurrent wind adjust the nominal ice thickness and concurrent wind pressure for Risk Category I structures from a 500-year mean recurrence interval to a 250-year mean recurrence interval. For Risk Category III and IV structures, they are adjusted to 1,000-year and 1,400-year mean recurrence intervals, respectively. The importance factor is multiplied times the ice thickness rather than the ice load because the ice load from Eq. (10.4-1) depends on the diameter of the circumscribing cylinder as well as the design ice thickness. The concurrent wind speed used with the nominal ice thickness is based on both the winds that occur during the freezing rainstorm and those that occur between the time the freezing rain stops and the time the temperature rises to above freezing. When the temperature rises above freezing, it is assumed that the ice melts enough to fall from the structure. In the colder northern regions, the ice generally stays on structures for a longer period of time after the end of a storm resulting in higher concurrent wind speeds. The results of the extreme value analysis show that the concurrent wind speed does not change significantly with mean recurrence interval. The lateral wind-on-ice load does, however, increase with mean recurrence interval because the ice thickness increases. The importance factors differ from those used for both the wind loads in Chapter 6 and the snow loads in Chapter 7 because the extreme value distribution used for the ice thickness is different from the distributions used to determine the extreme wind speeds in Chapter 6 and snow loads in Chapter 7. See also Table C10.4-1 and the discussion under Section C10.4.6.

C10.4.6 Design Ice Thickness for Freezing Rain. The design load on the structure is a product of the nominal design load and the load factors specified in Chapter 2. The load factors for load and resistance factor design (LRFD) for atmospheric icing are 1.0. Table C10.4-1 shows the multipliers on the 500-year mean recurrence interval ice thickness and concurrent wind speed used to adjust to other mean recurrence intervals.

The studies of ice accretion on which the maps are based indicate that the concurrent wind speed on ice does not increase with mean recurrence interval (see Section C10.4.4).

The 2002 Edition of ASCE 7 was the first edition to include atmospheric icing maps. Fifty-year mean recurrence interval (MRI) maps were provided at that time in order to match the approach used for the wind and snow load maps. An ice-thickness multiplier equal to 2 was included (see Eq. (10.4-5) in ASCE 7-02 to ASCE 7-10) to adjust the mapped values to a 500-year MRI for design. Thus, the adjusted 500-year MRI value would be appropriate for use with the LRFD load factor of 1.0 shown for ice loads in Chapter 2. A 500-year MRI load was consistent at that time with those historically used for seismic and wind loads. The 2016 edition of ASCE 7 includes atmospheric icing maps based on a 500-year MRI load with no ice-thickness multiplier in Eq. (10.4-5). The 2016 maps have been redrawn directly from the original extreme value analysis. Design load changes from the 2010 edition to the 2016 edition are caused by the maps being redrawn, not caused by changing the map MRI.

When the reliability of a system of structures or one interconnected structure of large extent is important, spatial effects should also be considered. All of the cellular telephone antenna structures that serve a state or a metropolitan area could be considered to be a system of structures. Long overhead electric transmission lines and communications lines are examples of large, interconnected structures. Figs. 10.4-2 through 10.4-6 are for ice loads appropriate for a single structure of small areal extent. Large, interconnected structures and systems of structures are hit by icing storms more frequently than a single structure. The frequency of occurrence increases with the area encompassed or the linear extent. To obtain equal risks of exceeding the design load in the same icing climate, the individual structures forming the system or the large, interconnected structure should be designed for a larger ice load than a single structure (CEATI 2003, 2005; Chouinard and Erfani 2006; Golikova 1982; Golikova et al. 1982; Jones 2010).

Table C10.4-1 Mean Recurrence Interval Factors

Mean Recurrence Interval	Multiplier on Ice Thickness	Multiplier on Wind Pressure
25	0.40	1.0
50	0.50	1.0
100	0.625	1.0
200	0.75	1.0
250	0.80	1.0
300	0.85	1.0
400	0.90	1.0
500	1.00	1.0
1,000	1.15	1.0
1,400	1.25	1.0

C10.5 WIND ON ICE-COVERED STRUCTURES

Ice accretions on structures change the structure's wind drag coefficients. The ice accretions tend to round sharp edges, reducing the drag coefficient for such members as angles and bars. Natural ice accretions can be irregular in shape with an uneven distribution of ice around the object on which the ice has accreted. The shape varies from storm to storm and from place to place within a storm. The actual projected area of a glaze ice accretion may be larger than that obtained by assuming a uniform ice thickness.

C10.5.5 Wind on Ice-Covered Guys and Cables. There are practically no published experimental data giving the force coefficients for ice-covered guys and cables. There have been many studies of the force coefficient for cylinders without ice. The force coefficient varies with the surface roughness and the Reynolds number. At subcritical Reynolds numbers, both smooth and rough cylinders have force coefficients of approximately 1.2, as do square sections with rounded edges (Fig. 4.5.5 in Simiu and Scanlan 1996). For a wide variety of stranded electrical transmission cables, the supercritical force coefficients are approximately 1.0 with subcritical values as high as 1.3 (Fig. 5-2 in Shan 1997). The transition from subcritical to supercritical depends on the surface characteristics and takes place over a wide range of Reynolds numbers. For the stranded cables described in Shan (1997), the range is from approximately 25,000 to 150,000. For a square section with rounded edges, the transition takes place at a Reynolds number of approximately 800,000 (White 1999). The concurrent 3-s gust wind speed in Figs. 10.4-2 through 10.4-5 for the contiguous 48 states varies from 30 to 60 mi/h (13.4 to 26.8 m/s), with speeds in Fig. 10.4-6 for Alaska up to 80 mi/h (35.8 m/s). Table C10.5-1 shows the Reynolds numbers (using U.S. standard atmosphere) for a range of iced guys and cables. In practice, the Reynolds numbers range from subcritical through critical to supercritical depending on the roughness of the ice accretion. Considering that

Table C10.5-1 Typical Reynolds Numbers for Iced Guys and Cables

Guy or Cable Diameter (in.)	Ice Thickness t (in.)	Importance Factor I_w	Design Ice Thickness t_d (in.)	Iced Diameter (in.)	Concurrent 3-s Gust Wind Speed (mi/h)	Reynolds Number
			Contiguous 48 States			
0.250	0.25	0.80	0.20	0.650	30	15,200
0.375	0.25	0.80	0.20	0.775	30	18,100
0.375	1.25	1.25	1.563	3.500	60	163,000
1.000	0.25	0.80	0.20	1.400	30	32,700
1.000	1.25	1.25	1.563	4.125	60	192,000
2.000	1.25	1.25	1.563	5.125	60	239,000
			Alaska			
0.250	0.25	0.80	0.20	0.650	50	27,000
2.000	0.50	1.25	0.625	3.250	80	202,000

Note: To convert in. to mm, multiply by 25.4. To convert mi/h to km/h, multiply by 1.6.

the shape of ice accretions is highly variable from relatively smooth cylindrical shapes to accretions with long icicles with projected areas greater than the equivalent radial thickness used in the maps, a single force coefficient of 1.2 has been chosen.

C10.6 DESIGN TEMPERATURES FOR FREEZING RAIN

Some ice-sensitive structures, particularly those using overhead cable systems, are also sensitive to changes in temperature. In some cases, the maximum load effect occurs around the melting point of ice (32°F or 0°C) and in others at the lowest temperature that occurs while the structure is loaded with ice. Figs. 10.6-1 and 10.6-2 show the low temperatures to be used for design in addition to the melting temperature of ice.

The freezing rain model described in Section C10.4.2 tracked the temperature during each modeled icing event. For each event, the minimum temperature that occurred with the maximum ice thickness was recorded. The minimum temperatures for all the freezing rain events used in the extreme value analysis of ice thickness were analyzed to determine the 10th percentile temperature at each superstation (i.e., the temperature that was exceeded during 90% of the extreme icing events). These temperatures were used to make the maps shown in Figs. 10.6-1 and 10.6-2. In areas where the temperature contours were close to the wind or ice thickness contours, they were moved to coincide with, first, the concurrent wind boundaries, and, second, the ice zone boundaries.

C10.7 PARTIAL LOADING

Variations in ice thickness caused by freezing rain on objects at a given elevation are small over distances of about 1,000 ft (300 m). Therefore, partial loading of a structure from freezing rain is usually not significant (Cluts and Angelos 1977).

In-cloud icing is more strongly affected by wind speed, thus partial loading caused by differences in exposure to in-cloud icing may be significant. Differences in ice thickness over several structures or components of a single structure are associated with differences in the exposure. The exposure is a function of shielding by other parts of the structure and by the upwind terrain.

Partial loading associated with ice shedding may be significant for snow or in-cloud ice accretions and for guyed structures when ice is shed from some guys before others.

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